CHAPTER I

INTRODUCTION

This chapter introduces Land Grand, a proposed educational game for Grade 5 learners at Baluan Elementary School. It explains the project context, purpose, objectives, scope, and limitations.

Project Context

This section explains the background and learning concerns that guide the development of Land Grand. It begins with educational technology worldwide, then discusses home learning in the Philippines and the needs reported at Baluan Elementary School.

Educational technology should respond to the needs of learners and teachers. UNESCO's 2023 Global Education Monitoring Report recommends choosing technology that fits local conditions and supports fair access to learning. It also stresses the need for evidence and resources that can be maintained over time (UNESCO, 2023). These points guide the proposed use of a learning game at Baluan Elementary School.

The value of a digital resource depends on how it supports the lesson. In Land Grand, short explanations would be linked to activities involving landforms, rocks, and minerals. Learners would explore a game environment, collect items, and read descriptions related to the lesson. The project would examine this approach as an additional learning resource rather than assume that playing a game guarantees better understanding.

In the Philippines, printed and digital modules were among the learning options used during the COVID-19 period. The Department of Education's 2020 guidance allowed supporting materials, including interactive electronic resources, alongside distance-learning activities. It also described the role of adults who assist learners at home (Department of Education, 2020, Appendix A and Appendix B). This experience provides a national background for developing resources that support learning beyond the classroom.

Disasters can also affect the arrangements needed for learning at home. DOST-PHIVOLCS recorded a magnitude 7.8 earthquake offshore Sarangani on June 8, 2026 (DOST-PHIVOLCS, 2026). As of October 2026, Baluan Elementary School holds face-to-face classes. Modular learning is considered an option if another disaster interrupts classroom instruction. Land Grand is therefore intended to support home study during regular schooling and possible disruptions.

An initial survey at Baluan Elementary School on July 7, 2026 gathered responses from 20 parents and 20 teachers. Parent responses covered Grades 4–6, while teacher responses covered Kindergarten and Grades 3–6. The parent group was available during the scheduled parents' orientation meeting. Respondent selection was reported as randomized.

Difficulty explaining module lessons was reported by many parents. In the full parent group, 85% reported difficulty explaining these lessons to their children. Similar concerns appeared among parents of Grade 5 learners. These responses support providing additional explanations within the Science topics covered by the project.

Teachers held mixed views about the effectiveness of modular learning. Some surveyed Grade 5 teachers did not consider it effective in helping learners understand lessons. Others were uncertain or considered it effective. These views support considering additional learning activities alongside printed materials, while keeping the role of teacher guidance.

The Grade 5 topic responses give a more specific basis for the proposed content. Some parents mentioned landforms among the Science topics their children found difficult. Other parent and teacher responses mentioned rocks or minerals. These are distinct content areas that Land Grand would cover through lessons, exploration, and item descriptions.

Most parents and teachers believed that a mobile game could support printed modules. However, some parents were unsure or unwilling to allow its use. Reported access to a smartphone, tablet, or computer also does not confirm access to a compatible Android device. These differences support offering the game as an additional resource while retaining printed materials and teacher guidance.

The survey records adult observations and views across several grades. The Grade 5 groups contain six parent responses and five teacher responses. These small groups help describe local concerns but cannot represent all Grade 5 learners. The findings also do not establish whether Land Grand improves knowledge, since the survey did not evaluate the game.

The survey identified the following four reported problems relevant to the proposed Land Grand project:

1. Some surveyed parents of Grade 5 learners reported difficulty explaining lessons in printed modules to their children.
2. Some surveyed Grade 5 teachers did not consider modular learning effective in helping learners understand lessons.
3. Some parents of Grade 5 learners reported landforms as a difficult Science topic for their children.
4. Some Grade 5 parent and teacher responses identified rocks or minerals as difficult Science topics.

These concerns support developing Land Grand while regular classes continue, before another disruption increases reliance on home study. The game would bring short lessons, item descriptions, and review activities into an offline resource. Learners could return to these activities across separate play sessions. This approach is intended to support home learning alongside the materials and guidance already provided by the school.

Purpose and Description of the Project

This section explains the purpose of Land Grand and describes its main learning activities. It shows how the proposed game would support Grade 5 learners alongside printed learning materials.

The game aims to support knowledge of landforms, rocks, and minerals through guided exploration. It would contain four episodes covering Mountains, Hills, Plains, and

Plateaus. Before each level, learners would read a short lesson about the landform and selected rocks and minerals.

During play, learners would move and jump with an explorer character in a two-dimensional environment. They would collect rocks and minerals and read each item's name and description. Obstacles and other challenges would form part of the exploration activities.

After a level, learners would see the collected items, a short review, and up to three gameplay stars. They could replay completed content or continue to the next available level. The game would save progress on the device for later sessions.

The project connects lesson explanations with exploration, collection, and review in one offline application. This design is intended to give learners another way to revisit the covered concepts. It would support printed materials and teacher guidance during home study.

Objectives of the Project

This section presents the general and specific objectives of Land Grand. The four specific objectives follow the same order as the reported problems and describe features that can be built and checked.

The general objective is to develop Land Grand, an offline Android exploration game for Grade 5 learners at Baluan Elementary School. It aims to support their knowledge of landforms, rocks, and minerals. Project completion is planned for November–December 2026.

The specific objectives are to complete the following within the planned period:

1. Provide a short Science lesson before each level to give Grade 5 learners additional explanations during home study.
2. Implement interactive exploration with movement and jumping controls and a review after each level to supplement printed Science learning materials.
3. Develop four episodes covering Mountains, Hills, Plains, and Plateaus, with lessons and exploration activities focused on these landforms.
4. Implement rock and mineral collection, with a pop-up showing each collected item's name and short description.

Scope and Limitations of the Project

This section describes the coverage and boundaries of the Land Grand project. It explains what the game includes and the limits of its use and the survey findings.

Scope

This section outlines the learners, content, game activities, and technical requirements included in Land Grand. Project completion is planned for November–December 2026.

The proposed game is for Grade 5 learners at Baluan Elementary School. Development would use Unity and C#, with Piskel and Clip Studio Paint for graphical assets. The application would include the following features:

- Four episodes covering Mountains, Hills, Plains, and Plateaus, with selected related rocks and minerals.

- Easy, Normal, and Hard modes with 3, 5, and 7 levels per episode, respectively. The proposed total is 60 levels across the four episodes.
- Short lessons before each level, item names and descriptions during collection, and a review after completion.
- A two-dimensional game environment with large touch buttons, background music, and sound effects.
- Movement, jumping, and item collection, with challenges such as obstacles, traps, pitfalls, and time limits.
- Progression through difficulty modes and episodes, with replay, continuation to available levels, episode summaries, and a final completion screen.
- Up to three gameplay stars after a level. Detailed star and completion rules would be specified during development.
- A Main Menu with Play, Settings, Tutorial, and Exit, and a Landform Map for selecting unlocked episodes.
- Automatic saving and loading of the player's name, unlocked episodes, completed levels, earned stars, and collected items on the device.
- Offline use on Android phones and tablets running Android 10 and above. The minimum targets are 2 GB RAM and 300 MB free storage, subject to device testing.

Limitations

This section explains the limits of the game and how the survey findings should be understood. It identifies the boundaries of the project's learners, content, and offline use.

- The intended users are Grade 5 learners at Baluan Elementary School. The survey includes other grades and cannot be treated as evidence about all Grade 5 learners.
- Learning content is limited to the four selected landforms and selected rocks and minerals. Detailed content and curriculum alignment still require validation.
- The game excludes multiplayer, online accounts, online features, cloud storage, websites, and an administrator panel.
- Saved progress stays on the device being used. The project does not include central teacher monitoring or shared learner records.
- The learning activities do not include separate quiz screens, GPS functions, or real-world scanning.
- The game supports printed materials and teacher guidance. Stars, collected items, and completed levels are gameplay records rather than measures of learning gains.
- The initial survey describes views and expectations about learning support. Any claim that Land Grand improves knowledge requires a defined evaluation method and actual results.